

RAPID RESPONSE

mNGS Training Playbook

*Metagenomic Next Generation Sequencing Training
for Pathogen Detection and Surveillance*

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How to Use This Playbook

This playbook provides a practical, end-to-end guide for planning, delivering, and sustaining Rapid Response (RR) metagenomic next-generation sequencing (mNGS) trainings. It is designed to support implementation of high-quality capacity-building activities and reflects lessons learned from multiple in-country and partner-hosted trainings. The playbook is modular and intended to be adapted to local contexts. Institutions should tailor timelines, workflows, and materials based on infrastructure, regulatory environments, and public health priorities.

Intended Audience

- **Public Health or Academic Institutions** planning to deliver mNGS training to support capacity building and the use of pathogen genomics for infectious disease surveillance.
- **Scientific leads and trainers** responsible for wet lab and dry lab instruction; providing scientific expertise and mentorship.
- **Project managers and coordinators** responsible for operations, planning, logistics, and timelines.

How This Playbook Is Organized

Six modules follow the full training lifecycle. Modules 3 and 4 include expanded day-by-day trainer guidance and are color-coded for quick navigation.

Module	Focus	Primary Audience
01 - Planning and Readiness	Establish partnerships, confirm scope, assess site readiness.	Program Director, Project Manager
02 - Pre-Training Logistics	Finalize structure, confirm trainees, prepare materials.	Project Manager, Wet Lab Lead, Analysis Lead
03 - Wet Lab Training Delivery	Deliver Week 1 laboratory sessions.	Wet Lab Lead
04 - Dry Lab Training Delivery	Deliver Week 2 data analysis sessions.	Analysis Lead
05 - Post-Training Support	Provide follow-up mentorship and implementation support.	Wet Lab Lead, Analysis Lead, Project Manager
06 - Evaluation and Continuous Improvement	Assess effectiveness and improve future trainings.	Project Manager, Program Director

Modules 3 and 4 include day-by-day session guides, QC checkpoints, and troubleshooting references designed for use by trainers in the field.

Each module follows a consistent structure:

- Module goals and timing
- Key decisions required during that phase
- Key activities and deliverables
- Roles and responsibilities
- Checklists, templates and resources

This structure is intended to help teams orient quickly, align expectations, and coordinate work effectively. Conducting a successful training involves many interdependent components. Early planning, clear communication, and strong organization are essential to delivering effective training and ensuring sustainable impact beyond the workshop itself.

There are templates and forms in Appendix A that are examples to support your planning, not requirements. They're organized by module — check the Module column, as a module may have several associated documents.

Note: This playbook is a living document. Teams are encouraged to download their own playbook and update content after each training based on their experience, evolving needs, and local context. When updating, please also revise the version number and date on the cover page.

Program Overview

Rapid Response (RR) mNGS trainings are designed to strengthen local and regional capacity for pathogen detection and genomic surveillance. The program addresses gaps in sequencing infrastructure, analysis expertise, and sustained technical support — particularly in settings with high infectious disease burden and limited diagnostic resources.

Training Model

Component	Description
Wet Lab Training	Study design, sample handling, nucleic acid extraction, library preparation, quality control, and ONT/Illumina sequencing fundamentals. (Week 1)
Dry Lab Training	Sequencing run QC, data interpretation, and metagenomic analysis using CZ ID (Week 2)
Implementation Support	Documentation, metadata standards, reproducibility, and troubleshooting in real-world settings.
Sustained Mentorship	Ongoing engagement as trainees implement workflows locally.

Team Roles and Responsibilities

Leads are responsible for ensuring all activities within their areas are completed – whether by doing them directly, delegating to teammates, or coordinating with partners.

Role	Primary Focus
Program Director	Strategic oversight, institutional alignment, funding coordination.
Wet Lab Training Lead	Laboratory training content and delivery. Designs and refines the wet-lab curriculum; oversees reagent and supply preparation; ensures equipment readiness; prepares and validates the training sample set; delivers lectures and leads hands-on lab activities; supervises lab setup, safety, and troubleshooting.
Analysis Training Lead	Post-sequencing data analysis and interpretation. Develops the analysis training curriculum focused on CZ ID; delivers analysis sessions; provides technical support during interpretation exercises; introduces

Role	Primary Focus
	genomic assembly, contamination assessment, and AMR detection tools; mentors trainees during case studies.
Project Manager / Research Coordinator	Operational execution and participant experience. Coordinates logistics for virtual and on-site components; manages timelines, scheduling, and communications; oversees regulatory documentation and compliance; manages reagent and sample shipping and documentation.

Communication and Coordination

The training team should operate with a culture of collaboration, trust, and adaptability. Team members are empowered to own their areas of responsibility, communicate proactively, and remain flexible — especially when working across institutions, regions, and time zones. Clear communication and consistent documentation are critical to running a successful training.

- **Slack:** Primary tool for day-to-day coordination, rapid problem-solving, and collaboration with international partners.
- **Email:** Used for formal decisions, communication with external partners, and documentation.
- **Zoom:** Used for planning sessions, decision-making meetings, and reviews.
- **Meetings:** Agenda-driven, time-limited, focused on decisions. Each meeting ends with documented action items and owners.
- **Shared project folder:** Single source of truth for agendas, timelines, decisions, templates, training materials, and other project documentation.

Anticipated Constraints and Adaptation

The training is designed with common implementation challenges in mind, including variability in reagent procurement and supply chains, differences in infrastructure and staffing, and local environmental or regulatory constraints. Training content and follow-up support are tailored based on site-specific assessments, with flexibility built in to accommodate adaptation while maintaining core quality, safety, and data integrity principles.

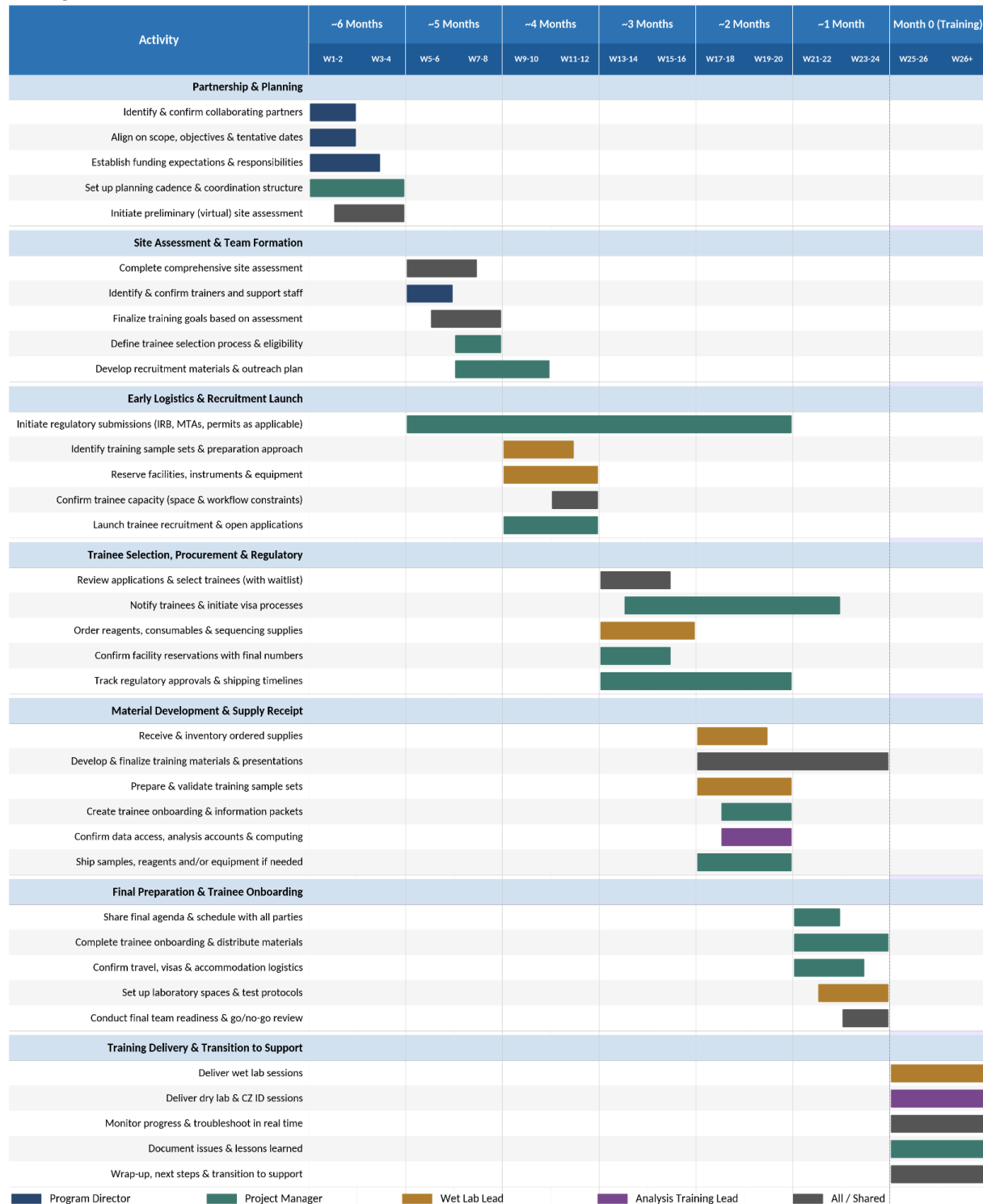
Overarching Goals

- Trainees are able to execute sequencing and analysis workflows in their home labs independently.
- Trainees can interpret sequencing and analysis outputs and communicate findings and limitations.
- Trainee home lab leadership establishes and maintains local continuity plans, including trained personnel, SOPs, and supply pathways.
- Trainees apply mNGS workflows to locally relevant surveillance, research, or outbreak response efforts.
- Trainees and trainers continue to engage in mentorship, knowledge exchange, and regional training activities.

Rather than focusing solely on one-time workshops, RR mNGS trainings emphasize long-term adoption and independent implementation. Trainings are paired with mentorship and follow-up support so participating laboratories can generate, analyze, and interpret sequencing data in alignment with local public health and research priorities. These goals above describe what a successful program looks like over time. For measurable, trainee-level indicators, see ‘Milestones to Track’ in Module 5.

Planning Timeline Overview

Planning Timeline — Gantt Chart



MODULE 1:

Planning and Readiness

Establish partnerships, confirm scope and assess site readiness.

M1	Planning and Readiness <i>Establish partnerships, confirm scope, and assess site readiness.</i>
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Goal	Establish partnerships, confirm scope and planning foundation, and assess site readiness.
Timing	~6 months before training

Key Decisions

- Proceed only if partner commitment, funding, and baseline laboratory feasibility are confirmed.
- Final training scope, objectives, and target cohort size agreed.
- High-level timeline and planning cadence established.

Key Activities and Deliverables

Identify and Confirm Partners and Team

- Identify and confirm collaborating partners.
- Identify and confirm trainers and support staff.
- Clarify roles, expectations, and levels of engagement.
- Draft and execute collaboration agreements.

Align on Scope and Objectives

- Define high-level training goals and intended outcomes.
- Identify target trainee profile and anticipated cohort size.
- Identify and initiate regulatory requirements to account for variable approval timelines

Establish Funding and Coordination

- Establish funding expectations and responsibilities. (see Appendix A)
- Set up communication channels (Slack, email, WhatsApp, recurring meetings).
- Establish a recurring planning cadence (e.g., monthly calls).
- Create and share the project folder (e.g., Google Drive). (see Appendix A)

Assess Site Readiness

- Conduct a preliminary (virtual) host laboratory assessment to evaluate space, equipment, personnel, and workflow capacity.
- Identify major gaps or risks requiring mitigation.

Establish Trainee Selection Process

- Draft eligibility criteria and application requirements.
- Define selection committee and decision timeline.

Documentation and Shared Workspace

Start at ~6 months; maintain through training.

- Create a shared project folder (Google Drive).
- Maintain preparation checklist and action tracker. (see Appendix A)
- Store regulatory documents and agreements for institutional records.

Budget & Cost Tracking

Start at ~6 months; maintain through training.

Use the list below to identify the categories that may need funding. Record estimated and actual costs in your institution's budget tracker spreadsheet once they are known.

- Training/Workshop Space or Room Rental: [venue rental, AV setup, cleaning/facilities fees, etc.]
- Reagents and Laboratory Supplies: [consumables, kits, shared equipment access, shipment]
- Shipping, Customs, and Import Duties: [customs fees, duty charges, and courier costs for imported reagents or equipment]
- Equipment Rental/Purchase: [lab or AV equipment rented or purchased for the training]
- Trainer & Staff Compensation/Stipends: [trainers, lab support staff, coordinators, guest lecturers]
- Trainee Housing or Accommodations: [hotel, dormitory, homestay]
- Food/Meals: [daily meals and coffee breaks]
 - Providing meals directly (rather than relying on trainees to source or cover their own) helps participants stay focused, arrive and return on time, and removes a financial barrier, especially for trainees traveling from other institutions or regions.
- Trainee Travel: [flights, ground transport to/from the airport]
- Visa/Immigration Fees: [application fees for trainees or visiting trainers, if covered by the program]
- Local Transportation: [shuttles, rideshare, parking, ground transport to/from the airport]
- Other Training-Related Logistics: [printing, signage, media/photography, translation, etc.]
- Internet/Connectivity Costs: [data plans, hotspot rental, especially for dry lab sessions]
- Contingency/Buffer: [a buffer for unplanned costs — commonly 10–15% of total budget]
- Total Budget/Funding Source: [amount and source]

Ideas for Reducing Costs

- Trainee/student housing (e.g., dormitories or shared housing instead of hotels)
- Existing institutional resources (space, equipment, or supplies already available)
- Other in-kind support (donated reagents, volunteer trainers, sponsorships)
- Partnering with the host institution to co-fund shared costs (space, equipment access)
- Partnerships with manufacturers/corporate partners
- Asking manufacturers for donated or discounted supplies/equipment

Post-Training Sustainability Costs

Beyond the workshop budget above, plan for the costs that sustain independent implementation after training ends:

- Reagent resupply for the first 6–12 months post-training
- Instrument service/maintenance contracts
- Funding line for cascade training (training additional personnel at the home institution)

Roles and Responsibilities

Role	Responsibility in This Module
Program Director	Leads partner identification; executes agreements; owns funding discussions; approves overall training scope and goals.
Wet Lab Lead	Participates in site assessment; identifies equipment, infrastructure and reagents requirements.
Analysis Training Lead	Reviews dry lab infrastructure needs; assesses data access and computing requirements.
Project Manager	Sets up shared folders and communication tools; maintains action tracker; schedules planning cadence.

Checklist: Module 1

- ☐ Collaborating partners identified and confirmed
- ☐ Trainers and support staff identified
- ☐ Collaboration agreement drafted and executed
- ☐ Training scope, goals, and target cohort size aligned
- ☐ Regulatory requirements identified and initiated (if applicable)
- ☐ Funding expectations confirmed
- ☐ Communication channels established (Slack, email, Zoom)
- ☐ Recurring planning cadence set
- ☐ Shared project folder created and shared
- ☐ Preliminary site assessment completed
- ☐ Gaps and risks from site assessment documented
- ☐ Trainee selection criteria drafted
- ☐ Working budget drafted and approved
- ☐ Catering, meals, and per diem responsibilities clarified

MODULE 2:

Pre-Training Logistics

Finalize training structure, confirm trainees, and prepare materials.

M2	Pre-Training Logistics <i>Finalize training structure, confirm trainees, and prepare materials.</i>
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Goal	Finalize all logistical, technical, and participant readiness requirements to ensure smooth training delivery.
Timing	~5 to 1 months before training (intensity increases closer to delivery)

Key Decisions

- Training dates confirmed; venue and lab space secured.
- Lead trainers and support staff confirmed.
- Training project and sample set confirmed.
- Regulatory/ethical pathway confirmed for the training sample set.
- Supplies ordered/received and equipment reservations confirmed.
- Final trainee cohort confirmed (including waitlist strategy) and communications sent.

Key Activities and Deliverables

Confirm Trainers, Venue, and Infrastructure (~5 months)

- Confirm trainers and support staff.
- Complete host lab readiness review (virtual walkthrough or photos) with gaps and mitigation plan.
- Confirm equipment, lab space, and trainee capacity based on space and workflow constraints.
- Confirm equipment and instrument reservation schedule.
- Finalize training goals based on assessment findings.

Study and Sample Set Preparation (~4–2 months)

- Decide on a training project and identify training sample sets.
- Confirm all regulatory/ethical requirements are approved or in progress.
- Ensure samples are extracted and aliquoted ahead of time if needed.
- Prepare sample metadata.
- If shipping is required, initiate shipment workflow (see Shipment section below).

Tip: Use a sample set you have run before and know works well for mNGS. Samples should be extracted and aliquoted prior to training if timing requires it. All samples must have appropriate IRB approval and consent. Complete the sample metadata form prior to the training. (see Appendix A)

Sample Strategy

- Strongly recommended: Use synthetic samples.
- Biobank samples: Only if well-stored, previously approved, and suitable for the planned workflow.
- If neither is available: Procure synthetic samples and add to the procurement timeline immediately.

Decision point: Are human samples being used? If yes, confirm existing IRB/ethical approvals are in place. Ensure MTAs are initiated and executed if materials are being transferred. Store all regulatory documents and agreements in the shared project folder. (see Appendix A for MTA templates)

Reagents, Consumables, and Equipment (~4–2 months)

- Review equipment and reagents list documents. (see Appendix A)
- Order reagents and confirm delivery timelines.
- Confirm recommended equipment list and reserve or procure anything missing.
- Maintain inventory tracking and storage plan.
- Have at least one extra sequencing kit available if feasible.

Sample and Reagent Shipment (~3–2 months, if applicable)

- Decide what will be shipped vs. sourced locally.
- Confirm sender/receiver points of contact.
- Confirm temperature requirements and packaging method.
- Confirm required approvals and documentation (MTA, permits, courier restrictions).
- Coordinate with courier; execute shipment.
- Confirm receipt, verification, and documentation at destination.

Tip: Assess local procurement conditions – shipping routes, customs timelines, local suppliers, and known delays. Target: all reagents received at least 2 months before the workshop. Order ~2x the reagents you anticipate needing in case of failures. Reserve equipment early.

Recruitment and Trainee Selection (~4–3 months)

- Define trainee selection process and eligibility criteria.
- Develop recruitment materials and outreach plan.
- Launch trainee recruitment and open application process.
- Review applications and select trainees (with waitlist).
- Notify trainees and initiate visa processes if needed.

Curriculum and Training Materials (~2–1 months)

- Create workshop schedule and daily flow. (see Appendix A)
- Prepare laboratory protocols.
- Develop workshop handouts, slide decks, and booklet.
- Compile CZ ID/computational training materials (datasets, exercises, accounts needed).
- Upload all materials to a shared Training Materials folder.
- Print workshop booklets and handouts if applicable.

The Illumina and ONT training booklets are platform-specific (wet lab protocols + CZ ID analysis worksheets). Reuse and update the existing versions each cycle rather than rebuilding them from scratch. (see Appendix A)

Trainee Onboarding (~3–1 months)

~3 months	Review applications; select trainees and waitlist; send acceptances.
~3–2 months	Collect trainee information forms; initiate visas if applicable.
~2–1 months	Share Visitor Guide; send expectations and preparation materials.

Roles and Responsibilities

Role	Responsibility in This Module
Program Director	Reviews training goals and scope; approves final cohort.
Wet Lab Lead	Orders and manages reagents and consumables; prepares and validates sample set; confirms equipment readiness; develops wet lab protocols and slide decks; conducts lab setup and protocol dry-run.
Analysis Training Lead	Develops CZ ID training materials and case study datasets; confirms trainee account access; prepares and loads backup datasets; reviews analysis exercises.
Project Manager	Manages trainee recruitment, selection, and communications; coordinates visa logistics; tracks and coordinates regulatory submissions; manages sample and reagent shipments; maintains master preparation checklist.

Checklist: Module 2

- ☐ Training dates confirmed and venue/lab space secured
- ☐ Lead trainers and support staff confirmed
- ☐ Host lab readiness review completed with gap documentation
- ☐ Training sample set identified and regulatory status confirmed
- ☐ Samples extracted, aliquoted, and metadata prepared
- ☐ Reagents and consumables ordered (target: received 2 months before training)
- ☐ Equipment reserved/procured; extra sequencing kit secured
- ☐ Shipment initiated (if applicable): approvals, courier, and documentation confirmed
- ☐ Trainee recruitment launched
- ☐ Trainees selected; acceptances sent; waitlist maintained
- ☐ Visa processes initiated (if applicable)
- ☐ Workshop schedule and daily flow finalized
- ☐ Slide decks, protocols, and handouts prepared
- ☐ CZ ID accounts and datasets confirmed
- ☐ Wet lab setup and dry run scheduled with Wet Lab Lead (see Module 3: Pre-Training Lab Setup)
- ☐ Dry lab setup and account/dataset readiness scheduled with Analysis Lead (see Module 4: Pre-Training Week 2 Setup)
- ☐ Training materials uploaded to shared folder
- ☐ Visitor guide and preparation materials sent to trainees
- ☐ Host lab point of contact confirmed for facilities and local logistics
- ☐ IT and computing readiness (internet connectivity) confirmed at venue
- ☐ Backup/contingency plan identified for critical equipment
- ☐ Reagents and samples received, verified, and confirmed in-lab
- ☐ Trainee information forms collected
- ☐ Final agenda and schedule shared with all trainees and partners
- ☐ Final logistics confirmed: travel/accommodation, facility reservation with final trainee numbers, catering/per diem, and emergency contacts/on-site safety briefing plan

MODULE 3:

Wet Lab Training Delivery

Hands-on laboratory sessions – Week 1

M3**Wet Lab Training***Hands-on laboratory sessions — Week 1.***FOR:** Wet Lab Lead + Supporting Scientists

Goal	Deliver hands-on wet lab training that builds genuine competency with the mNGS workflow from sample to sequence.
Timing	Week 1 of the training (Days 1–5, depending on cohort and site)
Primary Owner	Wet Lab Lead
Support	Support Scientist/Floater, Project Manager / Coordinator

Key Decisions Before Week 1 Begins

- Go/no-go confirmed for each major protocol phase before trainees proceed.
- Backup reagents and contingency plan confirmed for sequencing run failure.

Week 1 at a Glance

Sessions alternate between didactic instruction and hands-on benchwork. The schedule below is a guide — adapt timing to cohort size, local equipment, and sample type.

Day	Focus	Key Activities
Day 1	Orientation and study design	Program overview; introductions; mNGS introduction; study design principles; introduction to training sample set; lab safety and setup; bench assignments.
Day 2	Nucleic acid extraction	RNA and DNA extraction from training samples; homogenization if applicable; extraction QC (Qubit); group review of QC results.
Day 3	Library preparation — Part 1	Library prep overview (didactic); begin workflow: enzymatic steps, ligation; use incubation periods for Q&A and didactic sessions.

Day	Focus	Key Activities
Day 4	Library preparation — Part 2 and QC	Complete library prep; library QC (Qubit and Bioanalyzer/TapeStation); group review of QC results; troubleshoot any failures before proceeding.
Day 5	Sequencing run setup and launch	Pooling and normalization; flow cell loading (iSeq 100 or ONT); sequencing run launch; sequencing run QC overview while run is active.
	Buffer time/extension	Address protocol failures; repeat steps if needed; introduce additional wet lab topics: contamination controls, sample types, protocol adaptations for local context.

Day-by-Day Guidance

Pre-Training: Lab Setup and Dry Run

A trainer walk-through of the full protocol before trainees arrive is not optional — it catches equipment issues, reagent problems, and protocol gaps before they affect the training.

- Set up and organize bench space: reagent storage, pipettes, waste containers, PPE stations.
- Confirm all reagents are in stock, within expiration date, and stored at correct temperatures.
- Test all instruments: Qubit, Bioanalyzer or TapeStation, thermocyclers, centrifuges, sequencer.
- Confirm training samples are extracted, aliquoted, and labeled. Confirm sample metadata is complete.
- Brief all trainers and support team on cohort background, experience levels, and any language or accessibility needs.
- Print and distribute trainee workbooks and handouts.
- Set up a trainer communication channel for real-time coordination during the week.

Tip: Assign bench positions before Day 1. Group trainees by experience level if the cohort is mixed — pair less experienced trainees with more experienced peers for hands-on steps. Aim for a maximum of one trainer per five trainees so everyone gets adequate hands-on support.

Day 1: Orientation and Study Design

Morning — Didactic

- Program overview and introductions (trainer backgrounds, trainee introductions, institution context).
- mNGS introduction: what it is, why it matters, how it differs from targeted diagnostic approaches.
- Study design principles: sample types, controls, metadata, experimental design for infectious disease surveillance.
- Introduction to the training sample set: what the samples are, their context, and what results to expect.

Afternoon — Lab Orientation

- Lab safety walkthrough: PPE requirements, waste disposal, spill procedures, emergency exits.
- Equipment overview: brief demonstration of each instrument trainees will use.
- Bench assignment and setup.
- SPRI bead exercise: hands-on practice with magnetic-bead cleanup.
- Q&A: open discussion and questions about the week ahead.

Tip: Day 1 sets the tone. Create space for trainees to introduce themselves and share their lab backgrounds. Knowing who has hands-on sequencing experience and who is newer helps you calibrate pacing throughout the week.

Day 2: Nucleic Acid Extraction

Before starting:

- Review the extraction protocol with the full group (15–20 minutes).
- Demonstrate the first extraction step before trainees begin.
- Confirm all samples are at the bench, labeled, and at the correct temperature.

Hands-on:

- Trainees perform RNA and/or DNA extraction from training samples.
- Trainer and floater circulate to observe technique, answer questions, and catch errors early.
- QC checkpoint: Qubit quantification of all extracted nucleic acids.
- Review QC results as a group before ending the day. Flag any samples with low yield or quality concerns.

QC checkpoint: All samples should meet minimum concentration thresholds before proceeding to library prep. Samples that fail QC should be flagged for the group — either re-extracted or excluded with clear documentation.

Day 3: Library Preparation — Part 1

Before starting:

- Cover library prep overview didactically (30–45 min): what is happening to the molecules at each step, and why each step matters.
- Walk through the protocol on the board or slide before trainees touch pipettes.

Hands-on:

- Begin library prep workflow: fragmentation (if applicable), end repair, adapter ligation.
- Use incubation periods for Q&A, short didactic sessions, or breaks — not downtime.
- Confirm a clean stopping point by mid-afternoon to avoid running late.

Tip: Always confirm the stopping point before starting each step. Library prep has natural pause points(e.g., after ligation, before PCR amplification). Plan your day around these. Do not start a step that cannot be completed without running over.

Day 4: Library Preparation — Part 2 and QC

Hands-on:

- Complete library prep workflow: PCR amplification, cleanup.
- Library QC: Qubit quantification and Bioanalyzer or TapeStation fragment analysis.

Group review:

- Display QC results for the full group. Discuss: what does a good library look like? What do the peaks and concentrations mean?
- Identify any libraries that need troubleshooting before pooling.
- Troubleshoot failures with the group rather than bypassing them.

Day 5: Sequencing Run Setup and Launch

Before starting:

- Review pooling and normalization calculations with the group (20–30 minutes, didactic).
- Walk through sequencing run setup on the instrument before trainees begin.

Hands-on:

- Trainees perform pooling and normalization of libraries.
- Load flow cell (iSeq 100 or ONT MinION/Mk1C, depending on site instrument).
- Launch sequencing run.

While the run is active:

- Deliver sequencing run QC overview (didactic): what to look for in real-time metrics, how to interpret run quality.
- Review expected outputs and timelines for Week 2 analysis.
- Preview the Week 2 dry lab agenda. Allow time for informal questions and discussion.

Buffer Time (if the schedule allows)

Tip: This isn't a separate day — it's flexible time within the lab week. Use it to catch up if a step ran long or failed; if Week 1 stayed on track, use it for enrichment or hand the time back.

- Repeat any protocol steps that failed or where trainees need more practice.
- Troubleshoot the sequencing run if it hasn't completed cleanly.
- Cover additional wet lab topics based on cohort interest: contamination control, protocol adaptation for different sample types, local supply chain planning.
- Allow trainees to begin reviewing sequencing outputs once the run has completed.

Didactic Slide Decks Reference

All workshop materials for this module, including protocols, slidedecks and activities, are maintained in the [Rapid Response GitHub repository](https://github.com/cmtato/Rapid-Response-Resources), not in this playbook.

Repository URL: <https://github.com/cmtato/Rapid-Response-Resources>

Roles and Responsibilities — Week 1

Role	Focus During Wet Lab Week
Program Director	Attends opening and closing sessions; available as escalation point for institutional or partnership issues.
Wet Lab Training Lead	Leads all wet lab sessions; delivers didactic content; supervises hands-on benchwork; manages lab safety; makes go/no-go decisions at QC checkpoints; troubleshoots in real time.
Floater / Support Scientist	Circulates during hands-on sessions; provides 1:1 support to trainees who are struggling; assists with instrument operation and QC; flags issues to the Wet Lab Training Lead.
Project Manager / Coordinator	Manages daily schedule and logistics; ensures sessions run on time; documents issues, decisions, and action items; serves as primary point of contact for on-site issues.

Troubleshooting Guide — Wet Lab

Unexpected problems are not failures. When safe and feasible, work through issues with the group rather than around them. They are often the most valuable learning moments of the week.

Situation	Recommended Response
Low nucleic acid yield after extraction	Review technique with the group: homogenization, lysis efficiency, elution volume. Re-extract if time permits. If not, proceed with the best available sample and document.
Degraded RNA (low RIN / low quality)	Discuss RNA stability and cold-chain requirements with the group. If yield is sufficient, proceed and document.
Poor library QC — low yield or wrong size	Use as a teaching moment: review each step for failure points. Re-run PCR amplification if yield is low. Use backup library or pre-prepared library if QC is unacceptable.

Situation	Recommended Response
Sequencing run fails to start or shows poor metrics	Switch to backup dataset for Week 2. Treat run failure as a teaching moment for real-world troubleshooting. Document thoroughly.
Reagent shortage or expiration discovered mid-protocol	Use backup reagents if available. If not, adapt the protocol or use pre-prepped samples. Flag for procurement review after training.
Trainee falls significantly behind	Assign floater for 1:1 support. Have trainee share a bench station temporarily. Adjust overall pace if the issue is widespread.
Equipment malfunction	Activate contingency equipment if available. Escalate to host site contact immediately. Document for post-training report.

Checklists — Module 3

Pre-Training Lab Setup

- ☐ Lab setup and protocol dry-run completed before trainees arrive.
- ☐ Daily schedule confirmed and shared with all trainers and trainees before Day 1.
- ☐ Roles assigned for each session: who leads, who floats, who handles issues.
- ☐ Backup reagents and contingency plan confirmed for sequencing run failure.
- ☐ Bench space organized: reagents stored, pipettes calibrated, waste containers in place, PPE available
- ☐ All reagents in stock, within expiration date, and at correct storage temperature
- ☐ All instruments tested: Qubit, Bioanalyzer/TapeStation, thermocyclers, centrifuges, sequencer
- ☐ Training samples extracted, aliquoted, and labeled; sample metadata complete
- ☐ Trainee workbooks/handouts printed and distributed; bench assignments made
- ☐ Trainer communication channel set up; trainer briefing completed

Daily (Repeat Each Morning)

- ☐ Trainer check-in completed before trainees arrive (5–10 minutes)
- ☐ Session lead confirmed for each activity today
- ☐ Reagents and consumables for today staged at benches
- ☐ Issues from previous day reviewed and addressed

QC Checkpoints

- ☐ Day 2: Extracted nucleic acid concentrations meet minimum thresholds (Qubit)
- ☐ Day 4: Library concentrations and size distributions meet QC criteria (Qubit + Bioanalyzer/TapeStation)
- ☐ Day 5: Pooling and normalization calculations reviewed before loading
- ☐ Day 5: Sequencing run metrics confirmed within expected range after launch

End of Week 1

- ☐ Sequencing run launched and confirmed running, OR backup dataset confirmed ready for Week 2
- ☐ All wet lab QC data documented and saved to shared project folder
- ☐ Week 1 deviations and issues documented
- ☐ Trainer debrief completed and notes saved
- ☐ Dry lab trainer briefed and confirmed ready for Week 2
- ☐ Trainees briefed on Week 2 schedule and expectations

Tip: It's okay if things don't work out. A failed extraction, a low-yield library, or a messy run are real-world problems — work through them with the group rather than around them. These are often the most valuable teaching moments of the week.

MODULE 4:

Dry Lab Training Delivery

Data analysis and interpretation sessions – Week 2

M4**Dry Lab Training***Data analysis and interpretation sessions — Week 2.***FOR:** Analysis Training Lead + Supporting Scientists

Goal	Build trainee competency in sequencing data QC, metagenomic analysis using CZID, and interpretation and communication of results.
Timing	Week 2 of the training (Days 6–10)
Primary Owner	Analysis Training Lead
Support	Wet Lab Lead (for sequencing context), Project Manager / Coordinator

Key Decisions Before Week 2 Begins

- Go/no-go on data source: proceed with the Week 1 sequencing data, or switch to the backup dataset, based on run QC.
- Trainee readiness to proceed confirmed: all trainees can reach CZ ID on a working device, or a contingency (backup account, shared or paired devices) is in place.
- Analysis scope set: case-study depth matched to the cohort's level and the available data, including whether to include the optional AMR detection and genomic assembly content.

Week 2 at a Glance

Sessions build on each other: trainees first understand what the sequencer produced, then learn to analyze and interpret it, then practice communicating findings. The week culminates in a trainee presentation that functions as a capstone assessment.

Day	Focus	Key Activities
Day 6	Sequencing run QC and data handoff	Review sequencing run metrics; interpret run quality; understand what data was generated; transition from wet lab to analysis mindset; CZ ID account setup and orientation.

Day	Focus	Key Activities
Day 7	Introduction to CZ ID	CZ ID overview and navigation; uploading samples; understanding the pipeline; guided walkthrough of a complete sample analysis; hands-on exploration.
Day 8	Analysis and interpretation — Part 1	Filtering and reviewing mNGS outputs; distinguishing signal from noise; contamination assessment; interpreting positive and negative controls.
Day 9	Analysis and interpretation — Part 2	Case study exercises; trainees analyze their own results or assigned datasets; introduction to AMR detection and genomic assembly (if applicable); group discussion.
Day 10	Trainee presentations and wrap-up	Each trainee presents analysis findings; structured trainer feedback; wrap-up: next steps, post-training resources, support channels, post-training survey.

Day-by-Day Guidance

Day 6: Sequencing Run QC and Data Handoff

This session bridges Week 1 and Week 2. Trainees transition from thinking about library prep to thinking about data.

Morning — Didactic

- Review sequencing run QC metrics as a group: cluster density, Q30 scores, read counts, index assignment.
- Walk through what good vs. poor run metrics look like and what they indicate about the library or run.
- Introduce the data pipeline: what happens between raw reads and CZ ID output?

Afternoon — Hands-on

- Trainees log into CZ ID . Confirm all accounts are working before proceeding.
- Upload sequencing data from Week 1 (if run is complete), or introduce the backup dataset.
- Orient trainees to the CZ ID interface: projects, samples, runs, pipeline status.

If using the backup dataset: frame it positively. Explain that backup datasets are a standard part of training design and that analyzing pre-run data is exactly how trainees will troubleshoot runs at their home institutions when results are unexpected.

Day 7: Introduction to CZ ID

Morning — Guided Walkthrough

- CZ ID overview: what the platform does, how it processes data, what outputs it generates.
- Step-by-step guided walkthrough of a sample analysis: uploading, pipeline settings, reviewing outputs.
- Key outputs: taxon counts, coverage, background model, ERCC controls.

Afternoon — Hands-on Exploration

- Trainees explore their own CZ ID projects independently with trainer guidance available.
- Guided exercises: find the top hits, filter by taxon, look at coverage plots.
- Trainer circulates to observe navigation patterns and answer questions.

Tip: Give trainees unstructured time to explore CZ ID before introducing formal exercises. Letting them click around first builds familiarity and generates better questions.

Day 8: Analysis and Interpretation — Part 1

Morning — Didactic and Discussion

- Interpreting mNGS outputs: what does a hit mean? What does it not mean?
- Distinguishing signal from noise: understanding background, host reads, and low-complexity sequences.
- Contamination assessment: common contaminants in mNGS, how to use negative controls.
- The role of positive controls: expected results and what failures look like.

Afternoon — Hands-on

- Trainees apply contamination and QC assessment to their own dataset.
- Group discussion: what did you find? What concerns would you flag?
- Introduce the reporting framework: how do you communicate findings to a non-technical audience?

Day 9: Analysis and Interpretation — Part 2 and Case Studies

Morning — Additional Analysis Tools (adapt based on cohort goals)

- Genomic assembly in CZ ID: consensus genome and coverage visualization.
- AMR detection: overview of antimicrobial resistance gene identification in mNGS data.
- Pathogen identification confidence: when is a hit reportable? What additional confirmation is needed?

Afternoon — Case Studies

- Assign case study datasets (training samples, backup datasets, or pre-curated CZ ID public data).
- Trainees analyze their assigned dataset and prepare a brief summary of findings.
- Trainer circulates; encourage trainees to work through uncertainty rather than asking for answers.
- Close with group discussion: what did different trainees find? How did interpretation differ?

Case study design tip: Use datasets where the answer is not obvious. Datasets with a mix of expected pathogens, background noise, and one ambiguous hit generate the richest discussions and best prepare trainees for real-world analysis.

Day 10: Trainee Presentations and Wrap-Up

The trainee presentation is the most important session of Week 2. Do not treat it as optional or cut it for time.

Morning — Trainee Presentations

- Each trainee (or small group) presents: samples analyzed, findings, what they would report, and what questions remain.
- Presentations: 5–10 minutes each with 5 minutes for questions.
- Trainers provide structured feedback: what was strong, what could be more precise, what follow-up analysis would add confidence.

Afternoon — Wrap-Up Session

- Review post-training support resources: how to get help, who to contact, Slack channel.
- Distribute and complete post-training survey.
- Discuss what comes next: returning home, first independent run, implementation milestones.
- Q&A: open floor for any remaining questions.
- Informal close: networking time and trainer-trainee discussions.

Tip: The wrap-up is your opportunity to set realistic expectations for the transition home. Acknowledge that the first independent run is often the hardest and that support is available. Trainees who leave with a clear plan and a known contact are far more likely to implement what they have learned.

Dry Lab Training Materials Reference

All workshop materials for this module including slide decks, guides, and worksheets are maintained in the [Rapid Response GitHub repository](https://github.com/cmtato/Rapid-Response-Resources), not in this playbook.

Repository URL: <https://github.com/cmtato/Rapid-Response-Resources>

Disclaimer

- These training resources were developed to be used with the CZ ID platform.
- In October 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqToID (access at: <https://seqtoid.org/>).
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqToID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis.

Genomic Epidemiology (Advanced, Optional)

For institutions ready to build on core mNGS analysis, an optional advanced genomic epidemiology playbook is available in the [Rapid Response GitHub repository](https://github.com/cmtato/Rapid-Response-Resources). This training is typically offered after trainees have completed Module 4 and gained hands-on experience with CZ ID.

Repository URL: <https://github.com/cmtato/Rapid-Response-Resources>

Roles and Responsibilities — Week 2

Role	Focus During Dry Lab Week
Program Director	Attends Day 10 wrap-up as the closing session for the full two-week program; available as escalation point for institutional or partnership issues.
Analysis Training Lead	Leads all dry lab and CZ ID sessions; delivers didactic content; supervises hands-on analysis exercises; provides feedback on trainee presentations; decides on dataset substitution if needed.
Wet Lab Lead	Available as a resource for questions about the sequencing run, library QC, and data generation context. May co-lead the Day 6 sequencing run QC session.

Role	Focus During Dry Lab Week
Project Manager / Coordinator	Manages daily schedule; distributes post-training survey on Day 10; documents issues and action items.

Troubleshooting Guide — Dry Lab

Situation	Recommended Response
Sequencing data not ready for Day 6	Confirm backup dataset is loaded in CZ ID. Frame as normal training design. Use the time to discuss what to do when runs are delayed in practice.
Trainee CZ ID account not working	Have a spare pre-configured account available. Pair the trainee with a working account temporarily while the issue is resolved.
Internet connectivity issues	Download key CZ ID screenshots and pre-run export files as offline backups. Core analysis concepts can still be taught with static outputs.
Trainee is lost in the analysis workflow	Return to the guided walkthrough. Pair the trainee with a stronger peer for exercises. Assign additional 1:1 support time.
Case study results are ambiguous or unexpected	Lean into it. Ambiguous results are the best teaching material. Guide a discussion: what additional data would you want? What would you report and with what confidence?
Presentations are running over time	Set and enforce a strict time limit. If all presentations are running long, reduce per-trainee time rather than cutting the session.

Checklists — Module 4

Pre-Training Week 2 Setup

- ☐ Sequencing data from Week 1 is available, OR backup dataset is confirmed and loaded in CZ ID.
- ☐ All trainee CZ ID accounts are active, verified, and trainees have logged in at least once.
- ☐ Computing setup confirmed: trainees have working devices with internet access.
- ☐ Session schedule confirmed and shared with trainees and co-trainers.
- ☐ Case study exercises prepared and tailored to the training sample set or backup dataset.
- ☐ Sequencing data from Week 1 uploaded to CZ ID, OR backup dataset confirmed loaded and accessible
- ☐ All trainee CZ ID accounts active and verified (trainees have logged in at least once)
- ☐ Backup CZ ID account available for contingencies
- ☐ All trainee devices confirmed with internet access and CZ ID access
- ☐ Session slide decks reviewed and adapted to training sample set
- ☐ Case study datasets selected and pre-loaded in CZ ID
- ☐ Offline backups (screenshots or exports) downloaded in case of connectivity issues
- ☐ CZ ID worksheets and exercises printed or shared digitally
- ☐ Post-training survey prepared and ready to distribute on Day 10

Daily (Repeat Each Morning)

- ☐ Trainer check-in completed before trainees arrive
- ☐ CZ ID access confirmed for all trainees
- ☐ Session materials distributed or accessible
- ☐ Outstanding issues from previous day reviewed and addressed

End of Week 2

- ☐ Trainee presentations completed
- ☐ Post-training survey distributed and collected
- ☐ Post-training support channels confirmed with all trainees
- ☐ Trainees briefed on next steps and implementation milestones
- ☐ Analysis outputs and results saved and documented
- ☐ Week 2 deviations and issues documented
- ☐ Trainer debrief completed and notes saved to shared project folder

MODULE 5:

Post-Training Support

Provide follow-up mentorship and implementation support.

M5**Post-Training Support***Provide follow-up mentorship and implementation support.***Goal**

Ensure trainees can implement mNGS workflows independently at their home institutions and sustain that capability over time.

Timing

Immediately following training through ~6–12 months post-training

Key Decisions

Post-training support model confirmed before training ends — not after: who is responsible, what cadence, which channels. Mentorship assignments made: each trainee or site knows their primary point of contact. Thresholds established for when to escalate a technical issue vs. resolve asynchronously.

Key Activities**Immediate Post-Training (Weeks 1–2)**

- Send trainees a follow-up email summarizing key resources, contacts, and next steps.
- Confirm access to the shared training materials folder for all trainees.
- Distribute post-training assessment if not already completed (see Module 6).
- Schedule a first check-in call (target: 2–4 weeks post-training).
- Begin drafting the post-training report while details are fresh (finalized in Module 6 — see "Post-Training Report").
- Document remaining reagent inventory (what is left and where) and any equipment notes (instrument issues, borrowed or shared equipment).

Ongoing Mentorship (~1–6 months)

- Maintain a dedicated communication channel (e.g., Slack/ Whatapp) for trainees to ask questions and share updates.
- Hold regular check-in calls with trainees. Monthly is a reasonable default; adjust based on need.
- Assist with troubleshooting as needed when trainees attempt workflows independently.
- Review sequencing runs and provide feedback.
- Support adaptation of protocols to other sample types, or other study priorities.
- Encourage write-up and dissemination of results.

Milestones to Track

A training is only successful if trainees actually use what they have learned. Track progress against these indicators:

- First independent sequencing run completed at home institution.
- First CZ ID analysis completed independently.
- Trainees able to troubleshoot common issues without trainer support.
- SOPs or local protocols drafted or adapted.
- mNGS applied to a locally relevant surveillance or research question.
- An internal institutional champion is identified to sustain and expand the program.

Sustained independent implementation depends on continued investment beyond the training itself — reagent resupply, instrument maintenance, and support for cascade training. Confirm funding responsibility for these costs before committing to post-training mentorship goals.

When to Escalate

Escalate to a synchronous call or direct trainer involvement when:

- A trainee has attempted the workflow twice with persistent failures.
- There is a potential safety or compliance issue.
- Trainees are stuck on a conceptual gap rather than a technical one.

Roles and Responsibilities

Role	Post-Training Focus
Wet Lab Lead	Technical troubleshooting support; protocol adaptation guidance.
Analysis Training Lead	CZ ID and data analysis support; review of trainee outputs.
Project Manager / Coordinator	Manages check-in schedule; tracks milestone progress; coordinates follow-up communications.
Program Director	Strategic oversight; escalation point for partnership or institutional issues.

Checklist: Module 5

- ☐ Post-training follow-up email sent within 1 week
- ☐ Shared materials folder access confirmed for all trainees
- ☐ Post-training assessment distributed and collected
- ☐ First check-in call scheduled
- ☐ Mentorship assignments confirmed and communicated to trainees
- ☐ Slack channel (or equivalent) active with trainees added
- ☐ Implementation milestone tracker set up
- ☐ Regular check-ins are scheduled
- ☐ Appendices: attendance, equipment log, reagent usage.

MODULE 6:

Evaluation and Continuous Improvement

Assess training effectiveness and improve future trainings.

M6**Evaluation and Continuous Improvement***Assess training effectiveness and improve future trainings.***Goal**

Assess training effectiveness, document lessons learned, and improve future trainings.

Timing

Concurrent with Module 5; post-training report completed within 4–6 weeks of training

Key Decisions

How lessons learned get incorporated into the next training cycle.

Key Activities**Trainee Evaluation****Immediate (end of training):**

- Post-training assessment: trainee knowledge and confidence self-assessment, session-by-session ratings, open feedback.
- Informal trainer observations
- Course feedback survey: distribute a short Week 1 (Wet Lab) survey and a separate Week 2 (Dry Lab) survey to capture in-the-moment reactions while the training is fresh (see "Course Feedback Survey Topics" below).

Course Feedback Survey Topics

- Run two short surveys, one at the end of Week 1 (Wet Lab) and one at the end of Week 2 (Dry Lab) rather than a single combined survey. Keep each to 10–15 minutes and distribute on-site to maximize response rate (Week 2 pairs naturally with the Day 10 wrap-up session in Module 4). Adapt question wording to that week's content; the topics below apply to both surveys unless noted.
- **Respondent background:** institution/organization, country of residence, current role, and reason for attending.
- **Overall experience:** overall satisfaction rating for the week; whether the pace (too fast/slow/just right) and difficulty (too easy/hard/just right) were appropriate.
- **Content:** whether content met expectations and was relevant and useful; anticipated benefit to academic or professional development; any topics the trainee wished had been covered.
- **Presenter:** presenter knowledge/expertise, effectiveness engaging participants, and open feedback for presenters.

- **Format:** effectiveness of teaching methods used that week (lectures, guest speakers, group activities, discussions); usefulness of handouts/slides; most and least useful or favorite/least favorite activities; value of peer interaction.
- **Logistics:** satisfaction with pre-workshop communication (timing, location, requirements); whether schedule/duration felt appropriate; any additional materials or resources wanted; challenges encountered during the week.
- **Application and next steps:** how the trainee plans to apply the skills at their home institution, and any specific projects or initiatives they intend to use them for (Week 2 survey only).
- **Suggestions for improvement:** aspects of the workshop that could be improved; ideas for future topics.
- **Likelihood to recommend:** 1–5 likelihood to recommend similar workshops/trainings to colleagues.
- **Additional comments:** open-ended space for anything else the trainee wants to share.

Longer-term (3–6 months post-training):

- Check in with trainees: Have they run workflows independently? What barriers have they encountered?
- Sequencer procured and installed at the home institution, if not already available at training.
- Sequencing reagents received and in use; lab setup rated “Good” or better for independent mNGS, with no major equipment gaps.
- Libraries generated independently with consistent quality and no unresolved contamination issues.
- Additional personnel within or outside the lab trained by the original trainees (cascade training).
- Originally trained personnel retained on the project, with no unplanned departures.
- mNGS data analyzed independently using CZ ID or another tool of the lab’s choosing.
- Manuscript, poster, or conference presentation using mNGS data in progress or completed.
- Additional grant applications submitted to sustain or expand the work.
- Self-reported confidence in study design, sample collection, wet lab techniques, troubleshooting, and teaching others measurably improved since training.

Trainer Debrief and Documentation

Conduct a structured debrief with the full training team within 1–2 weeks of training completion. Cover:

- What worked well and should be preserved.
- What did not work and why.
- Specific protocol, schedule, or logistics changes to make next time.
- Any unresolved issues requiring follow-up.

Document the debrief in writing and store it in the shared project folder. Institutional memory depends on it.

Post-Training Report

The post-training report is the primary output of this module. Although it is finalized here, drafting should begin earlier, during Module 5's immediate post-training activities, while training details and QC results are still fresh. Trainers should complete within 4 weeks of training and share with relevant stakeholders. A complete report includes:

- Training overview: dates, location, cohort summary, trainers.
- Objectives and whether they were met.
- Summary of wet lab and dry lab sessions.
- QC and sequencing run results from the training sample set.
- Lessons learned and recommended changes.
- Potential challenges or blockers identified for setting up the platform.
- Outline of mentorship plan and next steps based on training outcome and trainee feedback.

Note on IRB / Ethical Approval for Evaluation Data: If course feedback or survey data may be used for activities beyond internal program improvement — such as research, publication, or external reporting — IRB or ethics approval is likely required. We recommend reaching out to your institution's IRB before designing or distributing any evaluation instruments. Requirements vary by institution and country.

Continuous Improvement

After each training, update your playbook to reflect what changed:

- Revise checklists, templates, or timelines based on experience.
- Add new troubleshooting scenarios encountered.
- Note any regulatory or procurement changes relevant to future trainings.

This playbook is only as useful as it is current. Updating it after each training is part of the program — not optional housekeeping.

Roles and Responsibilities

Role	Evaluation Focus
Wet Lab Lead	Contributes to trainer debrief; reviews wet lab QC data; inputs to post-training report.
Analysis Training Lead	Contributes to trainer debrief; reviews dry lab session outcomes; inputs to post-training report.
Project Manager / Coordinator	Owns post-training report timeline; distributes and collects surveys; consolidates evaluation data.
Program Director	Reviews and approves post-training report; ensures lessons learned inform future planning.

Checklist: Module 6

- ☐ Post-training survey distributed before trainees depart
- ☐ Survey responses collected and compiled
- ☐ Trainer debrief scheduled and completed within 2 weeks
- ☐ Debrief notes saved to shared project folder
- ☐ Post-training report drafted and circulated for review
- ☐ Post-training report finalized and shared with stakeholders
- ☐ Follow-up survey sent at 6 months post-training
- ☐ Playbook updated based on lessons learned
- ☐ Implementation milestone data reviewed and document

Appendices and Resources

The following appendices and templates support implementation across all modules.

Appendix A: Template and Resource Index

Resource	Module	Link
Folder Structure Template	1-5	PB01_Folder Structure Template
Training Program Master Tracker	1	PB02_Training Program Master Tracker Template
Training Plan Template	1-5	PB03_Training Plan Template
Laboratory Capacity Survey	1/2	PB04_Lab Capacity Survey Template
MTA Template (Biological Sample)	1/2	PB05_MTA (Biological Sample-BLANK)
MTA Template (Human Sample)	1	PB06_MTA (Human Sample-BLANK)
Master Preparation Checklist / Tracker	1/2	PB07_mNGS Training Playbook Master Tracker
Study Intake Form	2	PB08_Study Intake Form
Dry Lab Intake Form	2	PB09_Dry Lab Intake Form
Equipment and Reagent List	2/3	PB10_mNGS Training Equipment Reagents Materials List
Sample Shipping Metadata Template	2	PB11_Sample Shipping Metadata Template
Example of Course Description	2	PB12_Example of Course Description
Example Training Schedule (Facilitator)	2/3/4	PB13_Example Training Schedule_facilitator
Training Communication Plan Template	1/2	PB14_mNGS Training Communication Plan Template
Illumina Training Booklet	2/3/4	PB15_Illumina mNGS Training Booklet

Resource	Module	Link
ONT Training Booklet	2/3/4	PB16 ONT mNGS Training Booklet

Appendix B: Regulatory Compliance Reference

Regulatory requirements vary by country and institution. Confirm applicable requirements with your institution's legal, compliance, and ethics offices.

Requirement	When Required	Who Initiates
IRB / Ethics Approval	Any training involving human biological samples or data. May be required for conducting surveys and evaluations.	Program Director or PI at sending institution
Material Transfer Agreement (MTA)	Any transfer of biological materials between institutions	Program Director; legal/tech transfer offices at both institutions
Import / Export Permits	Cross-border transfer of biological materials or certain reagents	Project Manager; coordinated with courier and receiving institution

Appendix C: Lessons Learned Log

Update this section after each training. Review collectively before the next planning cycle begins.

Training / Date	What Worked Well	What to Change	Owner
[Name, Date]			
[Name, Date]			
[Name, Date]			

Appendix D: Glossary of Terms and Abbreviations

Terms are listed for readers across a range of technical backgrounds. Definitions are brief and program-oriented rather than exhaustive.

Term	Definition
AMR	Antimicrobial resistance; genetic markers indicating a pathogen's resistance to drugs.
Bioanalyzer / TapeStation	Instruments that assess nucleic acid or library fragment size and quality.
CZ ID	The cloud-based metagenomic analysis platform used in the dry lab. <i>(Formerly known as CZ ID / Chan Zuckerberg ID)</i>
Flow cell	The consumable on which sequencing reactions occur (e.g., on an iSeq or ONT device).
iSeq 100	A benchtop Illumina short-read sequencing instrument.
IRB	Institutional Review Board; body that reviews and approves research involving human subjects.
Library preparation	Converting extracted nucleic acids into a form that can be sequenced.
Metagenomics	Sequencing all genetic material in a sample rather than targeting a specific organism.
mNGS	Metagenomic next-generation sequencing; unbiased sequencing for pathogen detection and surveillance.
MTA	Material Transfer Agreement; governs transfer of biological materials between institutions.
ONT	Oxford Nanopore Technologies; long-read sequencing platform (e.g., MinION, Mk1C).
Pooling / normalization	Combining libraries at balanced concentrations before loading a sequencing run.

Term	Definition
QC	Quality control; checkpoints confirming samples, libraries, or runs meet defined criteria.
Qubit	A fluorometer used to quantify nucleic acid concentration.
RIN	RNA Integrity Number; a measure of RNA quality/degradation.
SOP	Standard Operating Procedure; a documented, repeatable protocol.
SPRI	Solid-phase reversible immobilization; magnetic-bead method for nucleic acid cleanup.

Appendix E: Program Risk Register

This register covers program-level risks—broader issues beyond the bench-level troubleshooting in Modules 3 and 4. Review it during planning (Module 1) and again before each training. Adjust the likelihood and impact ratings to fit your specific site.

Data governance needs special attention because this program crosses borders: sequencing data and metadata may fall under different countries' rules on data-sharing, privacy, and biosecurity. Before training starts, get written agreement on who owns the data, where it's stored, how long it's kept, and whether/how it can be shared with others or made public.

Risk	Likelihood	Impact	Mitigation	Owner
Trainer unavailable or withdraws	Medium	High	Identify a backup trainer early; cross-train a support scientist; document sessions so they can be led by another.	Program Director
Funding delay or shortfall	Medium	High	Confirm funding before the go/no-go decision; phase spending; maintain a contingency line.	Program Director
Customs / import-export hold on samples or reagents	High	High	Initiate permits and MTAs early, if applicable; source locally where feasible; build buffer into the timeline; confirm courier restrictions.	Project Manager

Risk	Likelihood	Impact	Mitigation	Owner
Reagent shortage, expiry, or supply-chain delay	Medium	High	Order ~2x anticipated need; target receipt 2 months ahead; keep a backup sequencing kit.	Wet Lab Lead
Sample degradation or loss	Medium	High	Extract, aliquot, and QC in advance; maintain cold chain; keep synthetic backup samples.	Wet Lab Lead
On-site instrument failure	Low	High	Run a full pre-training instrument check; confirm service/warranty; have a backup dataset ready.	Wet Lab Lead
Host lab infrastructure gaps (space, power, or instrument access)	Low	High	Conduct a preliminary virtual host lab assessment early; document gaps and a mitigation plan before confirming dates.	Wet Lab Lead
Loss of internet or CZ ID access	Medium	High	Download offline backups (screenshots/exports); prepare a backup CZ ID account; confirm connectivity in advance.	Analysis Lead
Sequencing run failure	Medium	Medium	Keep a validated backup dataset loaded in CZ ID; have an extra kit; treat failures as teaching moments.	Wet Lab Lead
Trainee attrition (visa, travel, availability)	Medium	Medium	Maintain a waitlist; initiate visa processes early; confirm attendance close to the date.	Project Manager
Regulatory / IRB approval delay	Medium	High	Identify and initiate requirements ~6 months out; track approval timelines actively.	Program Director / PM